preface

We truly live in exciting times! The rise of cloud-native technologies, starting some 10 years ago with Docker and Kubernetes, and the availability of cloud offerings that enable you to run large-scale applications based on a microservices architecture have changed the way we write and operate software.

I had the luck and pleasure of being part of that journey, starting in the container space in 2015 and then working in the Kubernetes space until 2021. There was one aspect of cloud native that stood out to me: given the dynamics of containers and function-as-a-service, if you don't have insights into what's going on in your system and aren't able to ask ad hoc questions about the state and trends, you're effectively driving a car blindfolded. When I changed teams in AWS to focus on observability, OpenTelemetry had just been formed, and the space was quickly developing. Now, at the time of publication, it's fair to say that observability has gone mainstream.

One thing that I only realized in hindsight was that what drew me to the observability space, besides the open source nature of the ecosystem around the Cloud Native Computing Foundation (CNCF) project, was the fact that observability is essentially an application area of data engineering. It's about generating, collecting, storing, and querying data, based on pipelines. Why do I point this out? Before I got into the world of containers, I spent more than a decade in data engineering, first in applied research and then in a start-up, where I got to apply the lessons learned, back in the "big data" days.

When the opportunity came to share what I had learned in the past 20 years, both in the data engineering and cloud-native spaces, in the context of providing a hands-